Accessible sonification of movement: A case in Swedish folk dance

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ABSTRACT

This study presents a sonification tool – *SonifyFOLK* – designed for intuitive access by musicians and dancers in their sonic explorations of movements in dance performances. It is implemented as a web-based application to facilitate accessible audio parameter mapping of movement data for non-experts, and applied and evaluated with Swedish folk musicians and dancers in their exploration of sonifying dance. *SonifyFOLK* is based on the WebAudioXML Sonification Toolkit and is designed within a group of artists and engineers using artistic goals as drivers for the sound design. The design addresses challenges of providing an accessible interface for mapping movement data to audio parameters, managing multi-dimensional data and creating audio mapping templates for a contextually grounded sound design. The evaluation documents a diversity of sonification outcomes, reflections by participants that imply curiosity for further work on sonification, as well as the importance of the immediacy of the both visual and acoustic feedback of parameter choices.

1. INTRODUCTION

All sounds are the result of a movement. When a violinist moves the bow, the strings start vibrating and we hear the sound of the violin. In dancing, some movements – the footsteps, for instance – are audible, but many movements cannot be heard. Sonification presents a possibility to map these movements to sound in order to make some of their aspects tangible and interpretable in the audio domain. As such, it is a great tool for exploring relationships between music and dance, but so far it has been a method that requires a certain degree of expertise to apply in a specific domain. This study aims at making the process of creating sonifications of dance movements more accessible for dancers and musicians, and describes an artistically driven design process of a sonification tool for Swedish folk music. The overarching aim is to investigate how sonified dance can be useful in pedagogic and artistic contexts as extensions to the traditional performance practices.

In Swedish folk music the link between dancing and playing is strong: music and dance forms are interrelated and dancing is to this day usually accompanied by live music. Folk music researchers have claimed that metric and rhythmic patterns in folk music should be understood in connection with the related dances [1–4].

With motion capture (MoCap), data from markers or sensors attached on performers' bodies can be recorded with high precision, and the MoCap data can be analysed using various high-level modelling of body movements. Such studies have demonstrated how dance movements embody rhythmic and metric music structures [5, 6], in Scandinavian folk music and dance with an emphasis on forms with asymmetric beat [7], such as Norwegian springar [8, 9] and Swedish polska [10, 11].

The real-time mapping of movement-related parameters to audio synthesis parameters for the purpose of sonification of movement characteristics has been explored in many contexts [12–15]. Due to the closeness of hearing to proprioception and kinesthetic awareness, translations of movement data into sound can offer additional paths for communicating embodied, expressive and somatic aspects of performance [16–18]. Furthermore, making a dance sound allows numerous kinds of sonic, expressive interactions between the sonic and the visual aspects of performance.

The possibilities for performers to contribute to such explorations remain limited as expert knowledge of data recording, processing and sonification is required. This threshold may be high for performers in genres based on live performance with traditional acoustic instruments. We present a novel application – *SonifyFOLK* – with the aim to provide an accessible way to explore sonification as a tool for artistic expression with movement-generated sounds in folk music. *SonifyFOLK* is a further development of WebAudioXML Sonification Toolkit (WAST) [19], a web-application that makes audio parameter mapping accessible to non-experts. The application prototype was developed in a user-centered design process with the first author (a folk musician), two experienced dancers and the second author, who is a web-software developer. These developments concern the visual representation of movement data in the graphical user interface, the pre-processing and selection of movement parameters, and the design of audio generators, aiming for contextually relevant dance sonifications. Following this prototyping process, *SonifyFOLK* was applied in workshops with music and dance students who were invited to create their own movement sonifications using the templates offered in the application prototype.

For the analysis presented in this paper we rely on our documentation from designing *SonifyFOLK*, as well as on feedback from workshop participants. We describe the prototype design in response to the challenges of provid-

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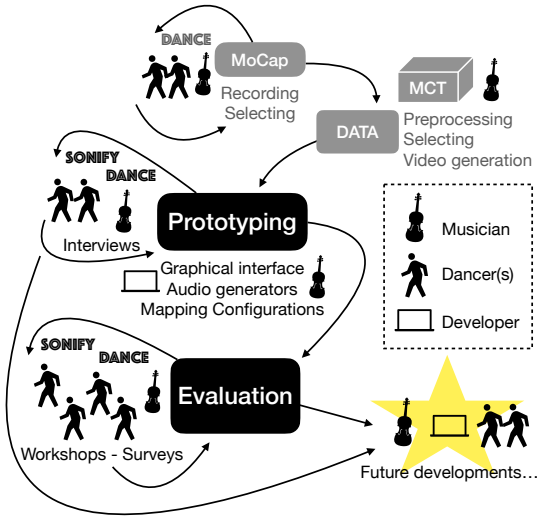


Figure 1. An overview of steps and actors in the described design and evaluation process.

ing performers with an accessible and interactive movement sonification tool. We then analyse the application of the prototype using survey responses and examples of mapping configurations devised by the participants in the workshops. Central findings are the importance of providing users with visual aids for accessible movements-to-audio mappings, as well as valuable examples, statements and reactions from dancers and musicians exploring the affordances of a sonification interface in their practice. Our results demonstrate how a user-centered design process involving dancers, musicians, and platform developer, using feedback from performing with sonifications, inspired a sound design responding to the performance context.

The paper is structured as follows: in Section 2 we refer to relevant research on movement and dance sonification and accessible sonification. Section 3 introduces the web-based technology supporting the application and the data preparation. In Section 4 we describe our methods for prototyping and evaluating the tool, Figure 1 provides an overview of this process. The results of evaluating the design are presented in Section 5 and further discussed in Section 6.

2. BACKGROUND

2.1 Movement sonification in various applications

Numerous studies on perception confirm advantages of the auditory domain over vision in information procession time, on time-based pattern and modulation discrimination, and describe cross-modal mapping between sound and body motion [15, 20, 21]. The close connection between auditory, motor, and vestibular systems allow sound to convey spatial proximity and temporal coincidence with the ability to stimulate proprioceptive and kinesthetic perception [18]. As such, goal-oriented interactive sonification of human movements has been explored with sensorimotor learning, sport and physiotherapy [12, 13, 22] and research tools have been suggested for gesture sonification in music-making for interpreting the expressive intentions of an instrumentalist [23].

2.2 Dance sonification

Sonification of dance involves steps of recording movement data, analysing and deriving relevant movement parameters, mapping to audio parameters and generating a sonic output, respectively. Depending on the research contexts and needs for interactivity, researchers have presented various approaches for movement data recording, including wearable, external, and optical sensors and video analysis. Methods for deriving movement parameters include various signal processing, motion analysis, gesture recognition, machine learning and periodicity extracting, and strategies for audio generation range from direct mappings of pitch and amplitude, the triggering of sound samples to modulating high-level musical parameters [8, 14, 15, 17, 22, 24–29].

Generating sounds from dance with MoCap has been explored in choreography and composition, for mediating expressive movement qualities and as a tool for learning dance movements [17, 30–32]. Drawing on emerging methodologies within dance practices that take advantage of sonification techniques to enhance corporal awareness, the term *Somatic sonification* has been proposed [18], as a meeting ground between data-driven scientific sonification and interactive music/dance application. Dance sonification has been applied in some contexts with traditional couple dances - i. e. in an ethnographic analysis of samba [29], and in the context of social dance, with interactive systems allowing Tango dancers agency over music. [33]. However, to the best of our knowledge, there have been no related studies within Scandinavian folk dance, albeit the strong connection between music and dance in these practices. Furthermore, the accessibility threshold remains at a high level for most of these systems.

2.3 Accessible sonification toolkits

There have been several attempts to build toolkits for for making sonification accessible to non-experts [34–36]. Commonly used platforms are Supercollider [37] and Pure data [38]. Since Web Audio API became a W3C recommendation, [1] a few online sonification toolkits have been presented [39–41] which makes sonification accessible in any web browser without any installations or specific platform requirements. This led to the development of WebAudioXML Sonification Toolbox (WAST) [19] that has been used in several courses by the author teaching sonification basics to non-experts.

3. TECHNICAL SETUP

This section explains the core technologies and the formation of the data-set used in the study. Section 3.1 and 3.2 explain how Web Audio API, WebAudioXML (WAXML) [42] and WAST are related and provide the foundation for *SonifyFOLK*. Section 3.3 describes the collection and preparation of MoCap data for sonification.

[1] Web Audio API, https://www.w3.org/TR/webaudio

3.1 WAXML

WAXML is an XML language and a javascript parser for simplifying the development of Web Audio applications. It is an abstraction of Web Audio API which is a high-level Web API for processing and synthesizing audio in web applications. The WAXML syntax includes XML elements for native Web Audio nodes and custom audio objects routing signals. It can be used for building audio generators, digital musical instruments, dynamic playback systems, and more. In this study, WAXML is used to build audio generators and configure the audio parameter mapping options. The language has been updated continuously since its first release and one of the most important new features made for this study is the support for named variables. It allows for a complex audio parameter mapping structure to be controlled using a single variable.

3.2 WAST

WAST [2] is a web-based graphical application built upon WAXML aimed at simplifying making sonification of CSV-data accessible without the need for any software installations and it has been chosen as the initial platform for the design of *SonifyFOLK* due to these characteristics. WAST is built with three main components: The data, the audio configuration, and the graphical user interface. The data is stored in a CSV-file in the same directory as the application and can be generated from any standard spreadsheet application. Every row represents a variable and every column represents a sample. The duration of the composition can be set independently of the number of samples. The audio configuration is stored within one or several linked WebAudioXML files. It typically contains a selection of audio generators. An audio generator can be viewed as a path with predefined variables that are exposed to the user. One such path can be built using oscillators, filters, audio loops, short samples, convolution reverb, etc, and exposes itself with a name and a set of possible variable targets for the user. The graphical interface consists of one graph displaying the data as a curve. The x-axis covers all samples and the y-axis is automatically adjusted to match the value-range of the data. It also has an interactive area where the user selects a variable and an audio generator and connects the variable to one or more variables in the audio generator. Finally, the mapping can be fine-tuned, inverted, or truncated using a set of sliders and check-boxes. WAST has since the first release been updated with better mixing features like binaural panning for better control over the final sound.

3.3 Motion captured performance data

The data for our study were collected from optical infra-red MoCap recordings of performances with two experienced Swedish folk dancers, Ami Dregelid and Andreas Berchtold, and one musician (identical to the first author). The recordings captured each dancer wearing a full body setup of 42 markers, the fiddle player wearing 19 body markers, and four markers attached to the violin and three to the bow. In order for the recording process to be ecologically valid and to capture a satisfying quality in the eye of the performers, performance recordings were iterated until a good capture was reached, with the participants rating their performance directly after each take.

For the prototyping of *SonifyFOLK* we selected the highest-rated recording of the tune *Polska efter Pellar Anna* related to the fiddle player *Gössa Anders Andersson* 1878-1963, from Orsa in Dalarna, Sweden. This tune belongs to a three-beat tune type - *polska* - characterized by asymmetric short-long-medium beat patterns [43], and it was chosen as one characteristic example of the Swedish folk music repertoire. The recorded couple dance contained two main alternating sections: (1) *promenad* with the dance characterized by walking side by side on a larger anti-clockwise circle, stepping on beat one and three, and (2) *omdansning*, where the couple rotates clockwise, holding and facing each other, completing one such couple rotation per measure [44]. During the 4.12 minutes of the performance the couple completed 7 transitions from *promenad* into *omdansning* and back to *promenad* again.

The MoCap data include three dimensional (xyz) position data for each marker. Using the MoCapToolbox for Matlab (MCT) [45], smoothing and gap-filling functions were applied, and with the function *mctimeeder*, the first derivative (*e.g.* velocity) was calculated in each dimension. Furthermore, using *mcnorm* the Euclidean norm of the vector data for each marker was computed, and, again using *mctimeeder*, the first derivative was derived from this single variable. This resulted in 8 dimensions for each of the 110 markers.

In order for the data to be manageable within the application, a subset of markers and dimensions had to be selected. This selection included three markers from each of the two dancers: one at each foot (ankles) and one from the upper back. Out of the 8 dimensions, the Euclidean norm of velocity was kept, and the position and velocity data of the vertical dimension. These choices were based on a previous study demonstrating such single marker movements as reliable visual cues for musicians [11]. In addition, the markers attached to the player's right toe and ankle, and to the frog of the violin bow were included, with the same set of dimensions. The two foot markers covered the musician's tapping of the beat, and the frog marker was included as an additional reference for movements connected with producing the music. With these reductions the dataset now included a total of 27 movement variables, with 9 markers in 3 dimensions. Finally, the frame rate was reduced to 24 fps in order to not overload the uploading of data through the web-interface.

4. METHOD

SonifyFOLK is designed and evaluated to meet two main goals. The first is to provide an accessible interface for dance sonification and the second is to offer synthesis models that would stimulate users to explore the artistic practice through sonification. The prototyping, described in Section 4.1 was done in collaboration with the two dancers that participated in the MoCap recording session. The eval-

[2] https://github.com/hanslindetorp/SonificationToolkit

uation, described in Section 4.2 was performed through workshops with music and dance students. The methods to collect data from this design study are described in each of these following sections.

4.1 Prototyping

The design strategy for this study is sequential [46, p. 24] and is based on a performance-led, iterative process with six steps that involves a musician (author 1), two dancers and a developer (author 2).

1. The musician creates an Audio Generator (a synthesis model using oscillators, audio files, filters, envelopes etc.) with WAXML
2. The musician selects one of the 27 movement variables and maps it to an audio parameter in the audio generator
3. The musician examines the artistic value of the mapping by playing to the sound
4. The dancers examine the artistic value of the mapping by dancing to the sound
5. The musician and the developer discuss new features to improve the mapping experience and the audio result
6. The developer adds new features to match the request and delivers a new version for the next iteration

The steps of the sequence were repeated during the prototyping in order to find audio configurations that are useful in the performance context, and to add features to the interface that facilitates a more accessible mapping of movement data to sound. The data from this user-centered design process was collected through the use of an annotated portfolio [47] including notes, emails, and annotated code. We also received the dancers perspectives through interviews and audiovisual recordings of rehearsals and performances.

4.2 Workshops

Following the prototyping of *SonifyFOLK*, evaluation workshops (2-3h) with music and dance students were conducted in three contexts: with students (3F) at the folk music department of the Royal College of Music in Stockholm, with folk dance students (3F, 1M) at the Department of Dance Pedagogy at Stockholm Uniarts, and with non-folk dance students (30F) at the same department. Participant ages were between 20 to 29 years and all provided informed consent.

The workshop started with all participants listening (the dancers also dancing) to examples of mapping configurations created by the first author. After being introduced to the *SonifyFOLK* interface, the participants created mapping configurations on their own devices using any combination of movement variables, audio generators, and mappings (not including the original fiddle recording).

This task was presented as to design sonifications "that I would like to dance to" (for the dancers) or "that are musically interesting" (for the musicians). Finally, the participants presented their sonifications, and the whole group was invited to dance to and further discuss the sonic outcome. The mapping configurations were collected using a survey that also included questions about their choice of mappings, what aspects of sound and movement they had found interesting with their sonification, and a possibility to suggest new features for further development of the application. The dance students were also asked to comment on their experiences from dancing to the sonifications, how it informed their understanding of the polska dance, and how they would rate the potential usefulness of movement sonifications in their own practice.

5. RESULTS

In Section 5.1, we present the results from the prototyping, organized in accordance with the two design goals presented in Section 4.1. In Section 5.2, we present the survey responses we got from the workshop evaluation with music and dance students.

5.1 Prototyping results

5.1.1 The graphical interface

The WAST interface displayed only static graphs of the data, but in order to enable a user to explore what movement variables to sonify we needed a simultaneous display of motion capture data and resulting sonification. Furthermore, the prototype should enable a user to relate selected movement variables to the full body movements of the performers, to navigate between different parts of the performance recording, and to receive a sonic output corresponding with the tempo of the performance.

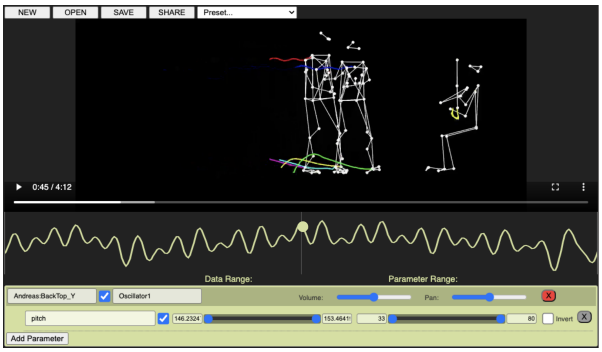


Figure 2. The graphic interface of the prototype with MoCap video, data graph display and area for selecting data variable and audio generator, adding audio mapping parameters, and sliders and check-boxes for adjusting mappings and sonic output.

The implemented solution is to synchronize sonification playback with a video showing stick-figure animations of the performers rendered from the MoCap-data (Figure 2), and adding audio of the music recorded during the performance. Since the selected movement variables (see Sec-

tion 3.3) contain mainly oscillating shapes of the repetitive dance and tapping movements, we added automatic scrolling of the data graphs during playback and zooming functions. In addition, we included dots moving vertically to the y-axis position of the data sample currently being mapped - as a playhead for the sonification. These bouncing dots, inspired by earlier findings [11], were intended to make it easier to follow the data parameters in real-time. During the design of the prototype, the simultaneous visual and sonic representations of dance proved valuable for trying out mappings, and for connecting sounds with specific movements in different sections of the performance.

5.1.2 *Audio synthesis for folk dance*

Sonifying dance movements could be seen as connecting the movement of a dancer to the movements when playing an instrument, thus closing a circle from sound to movement and back to sound. With the intention to promote co-performance, this inspired the design of audio generators with regards to tonal, timbral and rhythmic qualities. [3]

One example of audio generator design took advantage of using midi values as a grid for the step-wise mapping of incoming values to a pitch frequency parameter of an oscillator, as illustrated in Fig. 3. Pitches were then specified with floating point numbers in accordance with characteristic microtonal intonations of Swedish folk music. This feature allowed musicians to improvise with micro-intonation following such tonal sequences generated by the dance.

```
<var name="freq" value="$movement_variable_1"
  mapin="0,1"
  mapout="62, 74"
  steps="0, 2, 3.5, 5, 7, 8.5, 10, 12"
  convert="MIDI->frequency" />

<OscillatorNode frequency="$freq" />
```

Listing 1. Code example for how WAXML can map a movement variable to control the frequency of an ocillator according to a D scale where the third and sixth note is in between a minor and major interval.

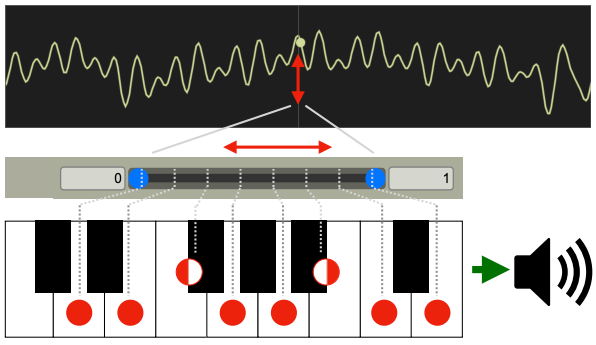


Figure 3. Illustration of the code example in Listing 1 showing how a value of a movement variable is mapped to a D scale with microtonal intonations.

[3] The preset menu at https://omisgeld.github.io/SonifyFOLKSMC2023 exemplify configurations with the described audio generators, and additional examples of configurations by workshop participants, as described in Section 5.2.

Further examples of audio generators included continuous sounds implemented by oscillators and looped samples, comparable with the drone-like sounds that are a common feature in much folk music. By mapping to signal amplitude and bandpass filter center frequency parameters these continuous sounds can be related to changes of e.g. bow speed and pressure by a fiddler. Others, with white-noise audio generators using similar mappings, create non-tonal, wind-like sounds, as metaphors for bodies moving through the air.

In addition to the synthesised sounds of oscillators and noise generators, additional audio generators included samples of fiddles in order to infer timbral qualities of a traditional instrument. For one such audio generator, a step-wise mapping of data to midi notes was set to filter out partials in looped samples of long fiddle notes. The resulting sound shifted between different harmonics, resembling playing flageolets on a fiddle, and with tonal references to overtone flutes or jaw harps, see Fig. 4.

```
<var name="freq" value="$movement_variable_1"
  mapout="69,76,81,84.86,88,90.69,93,95,96.86,
  98.51,100,101.41,102.7,103.88,105"
  curve="step" convert="MIDI->frequency" />

<AmbientAudio src="violin_loop.mp3"
  output="next" />

<BiquadFilterNode
  type="bandpass" frequency="$freq" Q="50" />
```

Listing 2. Code example to illustrate how movement variable can be mapped to control a bandpass filter stepping a series of harmonics using WAXML.

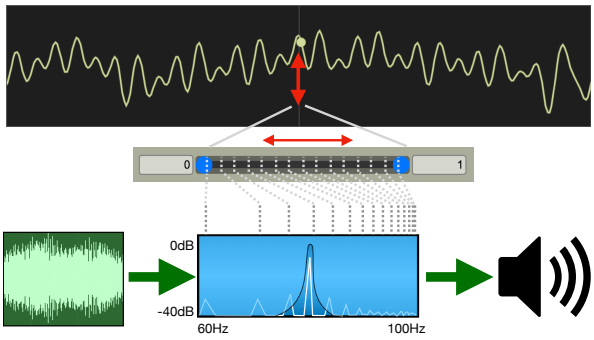


Figure 4. Illustration of the code example in listing 2 showing how a value of a movement variable is mapped to control a bandpass filter stepping a series of harmonics.

Apart from these continuous sounds, we explored using discrete sounds *e.g.* at the maxima and minima vertical motion of the center of gravity of the dancers. We added audio generators where we let crests and troughs trigger the attack of a short-envelope, percussive sound. This facilitated sonifying the foot-tapping of the player at the foot's maximum downwards velocity, in order for getting a reference for the beat. In our examinations, we observed the large extent to which such foot-tapping sonifications sounded simultaneous to similarly generated sonifications of the vertical velocity of the dancers' upper back markers. Adding parameters for attack and decay time facilitated the

data variables affecting the articulations of tone onsets, i.e. making grave or acute accents depending on the oscillations of movements as illustrated in Fig. 5. This feature added a musically relevant possibility - to create sounds with various types of attacks, a significant aspect of dance music performance [7].

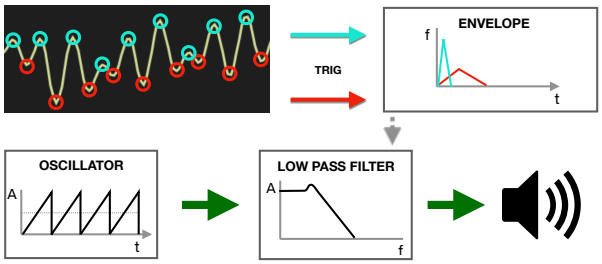


Figure 5. Illustration of how the crests and trought in the vertical movement variable is triggering an envelope that controls a low pass filter.

When asked about the continuous sounds the dancers reported that dancing to these sounds felt intuitive, enjoyable, and easy. They expressed that the sounds connected to their moving bodies, without the need of knowing what movements were currently being sonified, and invited them to emphasise certain movement aspects, for instance rotational, turning movements. With the more percussive sounds, the dancers reported that the sonifications were generally harder to follow, and that they became more occupied with understanding the metric alignment of the sounds. The dancers also stated that the attack and decay times of the sounds would influence their timing. In contrast to the dancers, the musician expressed the somewhat reversed preference: that it was easier to find the beat while playing to the percussive than to the continuous sounds. For this reason, both continuous and percussive sounds were included as possibilities for the workshop evaluations.

5.2 Workshop evaluation

The workshop survey collected 39 answers with mapping configurations devised by participants, and 17 responses to the additional questions for dancers. The 39 mapping configurations revealed a diversity of mappings using movement variables from all the provided 9 markers. Similarly, mappings to all available audio generators appeared at some instances of the mapping configurations. The majority of these configurations included mapping from two to three movement variables, many used only one variable, others up to a maximum of 7 simultaneous data sources. The most common movement variable to include was the vertical position of the upper back marker of one dancer. Another common choice was the left foot of one dancer. This distribution can be a result of the order that data variables appeared in the interface. Still, the bow marker, which was kept further down the list, was another popular choice.

In the surveys, participants commented that they recognised the dancing through the sonifications they created, finding that specific movements could be described through sounds. Participants also commented on the results of their mappings in musical terms such as ”groovy” and ”rhythmic”.

Dancers reported that dancing with the sonifications was both confusing, interesting and thought-provoking: “It was hard in the beginning to find the sound and match it to movements” (P4). After being introduced to the interface and the sources of sounds, dancers described getting further insights into the polska dance: “I could see the dynamics in the dance in a completely different way” (P12). Participants expressed that, in addition to focusing on specific dance movements, they were inspired to contemplate the relation between music and dance, “I get understanding for the body in motion and how it can be coupled to music with yet another layer” (P2). All survey responses included high ratings (3 or more on a 1 to 5 scale) for the potential usability of sonification in their own dance practice.

When asked to address potential improvements, participants mentioned that they had sometimes encountered problems with the stability of the interface, that they had trouble understanding the effects of certain movements and audio parameters while trying to find the sonic output they were searching for. Furthermore, participants suggested extended cross-mapping possibilities between movement variables and audio generators, for switching between alternative sonifications, and for allowing separate mappings in different parts of the performance.

6. DISCUSSION

The evaluation of the prototype provided evidence for the portability and accessibility of *SonifyFOLK*: participants were able to navigate the interface on their own devices, explore various mappings of movements to audio, and share their created configurations by simply posting a link. Based on the success with the interface, as described in Section 5.1.1, we draw the conclusion that the combination of synchronized video, audio, and animated data is an important feature for sonification of movements.

The development of *SonifyFOLK* was guided by the intention to allow artists to explore sonification in connection with performing. Performers' feedback in the prototyping gave valuable insights into the affordances of dancing and playing with different types of sonification designs. In the workshops, the participating musicians and dancers used the tool in various ways: creating sound designs, dancing to sonifications, focusing on specific aspects of dance and music performance. The responses collected during such interactions are valuable for understanding the utility of the tool in their artistic practice.

SonifyFOLK is currently limited to sonifying pre-formated performance data. Adding accessible solutions for uploading or streaming data from portable motion tracking devices would increase the possibilities for interactive approaches. Furthermore, including dynamic data processing possibilities within the interface, such as calculating speed and velocity, would allow an improved navigation of data sources and movement dimensions, and subsequently limit the need for pre-processing of the data. To these ends, exploring sonification of movement data captured using simple low-end sensors could be useful for evaluating more

accessible and mobile solutions. The preset audio generator configuration strongly impacts the available choices for the users and we can see a potential for having a plugin system similar to Digital Audio Workstations (DAWs) where a virtual instrument can be added dynamically. As many musicians and performers are accustomed to using DAWs, an interface that looked more like a software synthesizer, would potentially enhance the users sense of agency over the sonification output.

This study lays the foundation for further exploration of movement sonification with Swedish folk music and dance. Through the chain of recording a performance, extracting movement data, creating audio mappings and performing with movement sonification we see potentials for further studies on music and dance interactions. Although folk music relies on oral transmission, recorded sources influence how traditions are transmitted, re-shaped and re-articulated between generations of musicians and dancers. Movement sonification offers intriguing new perspectives for interacting with performance documentation and for generating new artistic output with folk music and dance. As such, the study demonstrates how artistic processes can assist in the design of interactive media technology, and how in turn this technology can extend the expressive horizons of an artistic practice.

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